

Nearest Neighbors

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Apply neighbour and tie rules for Nearest neighbours and distances. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Nearest neighbours and distances. Proposed explanation: “A negative difference is a negative distance”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Training points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute distance; for a nearest-neighbor tie choose the alphabetically earlier ID. Compare $k=1$ and $k=3$.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Training lengths 2(H),6(S),10(H); query=8. Compute distances. Explain the tie and why combining length in mm with width in cm changes a two-feature distance.

6. Solve the given case

One-dimensional training points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute distance.

Find $k=1$ neighbours and handle ties.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-neighbor with absolute distance, predict its label.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 12 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, scaling, distance, and k.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Training lengths 2(H),6(S),10(H); query=8. Compute distances. Explain the tie and why combining length in mm with width in cm changes a two-feature distance.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate feature distances

For query 3 cm, calculate distances to all five training rows. Explain why A and B tie even though their labels differ.

My reasoning and evidence

13. W2 Apply neighbour and tie rules

For query 3, list the selected IDs and predictions for $k=1$ and $k=3$ using alphabetical ID order on equal distances. Explain what changes and what remains the same.

My reasoning and evidence

14. W3 Keep evaluation examples separate

Why must the three validation reference labels not enter the neighbour pool? Give the fixed $k=3$ score and baseline score on the worked validation examples.

My reasoning and evidence

15. W4 Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with $k=1$ and $k=3$. Using the same training pool and $k=3$, score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

My reasoning and evidence

16. W5 Interpret a vote and a match

Two of three neighbours vote clear. Does that prove a two-thirds chance of a correct prediction? Does an exact feature match guarantee a correct label? Explain both limits.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

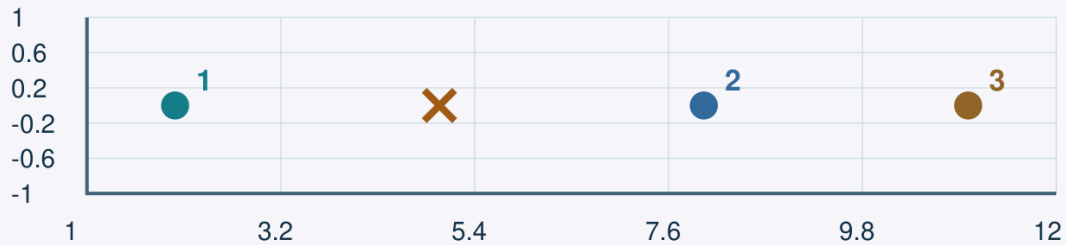
18. Pictorial case practice: Nearest Neighbors

Let a nearest-neighbour picture show each distance before voting.

Training: A length 2 healthy; B length 8 spotted; C length 11 spotted. Query length 5 cm. Distance is absolute difference; ties use alphabetically earlier ID.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, $y=0$

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

To A: $|5-2|=3$; B: $|5-8|=3$; C: $|5-11|=6$ cm.



2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.



3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length7 cm.

Explain the changed result using the matching lesson.

Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Let a nearest-neighbour picture show each distance before voting.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? All three vote: two spotted and one healthy, so $k=3$ predicts spotted.

1. Measure distances
2. Compare $k=3$
3. Apply $k=1$ rule

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare $k=3$ All three vote: two spotted and one healthy, so $k=3$ predicts spotted. Read the picture in order. All three vote: two spotted and one healthy, so $k=3$ predicts spotted. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Move the query to length7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. $k=1$ predicts spotted from B, distance1; $k=3$ remains spotted.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: $k=1$ predicts spotted from B, distance 1; $k=3$ remains spotted. Distances become 5, 1, 4 cm, so the nearest tie disappears.

Repair a missing building block

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf predictions from training examples. Short practice: Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
